St. Francis Institute of Technology

(Engineering College)

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Department of Artificial Intelligence and Machine Learning

Academic Year: 2025-2026 **Term**: Even (Jan. 2026 – Jun. 2026)
Class / Branch: SE – AIML **Semester**: IV
Course: Web Programming Lab. (AI4VS_LR4)
Date of Assignment: / /2026 **Date of Submission:** / /2026

Pre-Lab Exercises for Experiment-10

Exercise 1: Understanding Node.js Environment

Task:

1. Install Node.js.
2. Verify installation using:
   - node -v
   - npm -v
3. Create a folder named practice-node.
4. Initialize npm inside it.
5. Identify the purpose of and explain :
   - package.json
   - node_modules
   - dependencies

Exercise 2: Create a Basic Node Server (Without Express)

Task:

1. Create a simple HTTP server using Node's built-in http module.
2. Make it respond with:
   - "This is a Node server"
3. Run the server.
4. Access it in the browser using localhost.

Observation:

- What happens when you refresh the browser?
- What happens if you change the port?

Exercise 3: Install and Explore Express

Task:

1. Create a new folder express-practice.
2. Initialize npm.
3. Install Express.
4. Observe and explain
   - Changes in package.json

   - Creation of node_modules
5. Write down:
   - What is Express?
   - Why is it used?

Exercise 4: Create Simple Routes

Task:

1. Create an Express server.
2. Create the following routes:
   - / → Send text response
   - /contact → Send text response
   - /help → Send text response
3. Test each route in browser.

Questions:

- What happens if you type a route that is not defined?
- How does Express differentiate routes?

Exercise 5: Return JSON Data

Task:

1. Create a route /api/user.
2. Return JSON data with:
   - name
   - age
   - course
3. Test using:
   - Browser
   - Postman

Questions:

- What is the format of JSON?
- How is JSON different from plain text?

Exercise 6: Middleware Practice

Task:

1. Create middleware that logs:
   - Request method
   - Request URL

2. Attach it to the application.
3. Access different routes.

Observation:

- Does middleware run for every request?
- When does it execute?

Exercise 7: nodemon Practice

Task:

1. Install nodemon.
2. Run your Express server using nodemon.
3. Modify your server file.
4. Observe behavior.

Questions:

- What is the difference between running with node and nodemon?
- Why is nodemon useful during development?

Exercise 8: 404 Handling

Task:

1. Create a handler for undefined routes.
2. Return:
   - "Page Not Found"
3. Test with an invalid URL.

Question:

- Why is 404 handling important in backend development?

Exercise 9: Request–Response Analysis

After completing all exercises, write:

1. Step-by-step explanation of what happens when:
   - A user types localhost:3000/ in browser.
2. Explain:
   - What is a request?
   - What is a response?
   - Where does middleware execute?